BACKEND · CROSS-BORDER COMMERCE · CHONGQING, CHINA

Xiaozhe Bai

Backend Software Engineer focused on cross-border e-commerce systems.

jeff2025white@gmail.com Upwork Intro video

01 ABOUT

Backend engineer with 7 years at Alibaba, building distributed systems for AliExpress's global cross-border business. The work centers on order and transaction systems, merchant open APIs, high-concurrency optimization, and multi-region deployment — keeping high-traffic services stable through performance tuning and fault isolation.

02 SKILLS

Java · Go · Microservices · MySQL · Redis · Kafka · RocketMQ · API gateway · Distributed locks · DB sharding · Python · FastAPI · PostgreSQL · LLM integration · OCR / computer vision · PHP · Laravel · Android

03 EXPERIENCE

Backend engineer

Alibaba · AliExpress · 7 years

Owned backend services for order, settlement, and merchant open APIs on a global cross-border marketplace, from high-concurrency tuning to multi-region deployment. See order & settlement and open API gateway below.

04 EDUCATION & LANGUAGES

Shanghai Jiao Tong University

Bachelor of Applied Science (BASc), Computer Science · 2015 – 2019

Languages

Chinese — native or bilingual

English — fluent (written)

EMPLOYER WORK

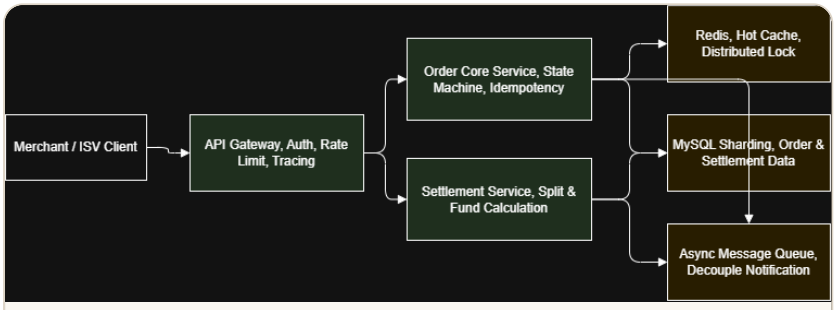
Selected work from a previous employer. Descriptions stay high-level and do not disclose confidential business data, internal metrics, or proprietary code.

Cross-border Order & Settlement Backend

Backend Software Engineer · AliExpress / Alibaba

- Owned the order/settlement pipeline for a global cross-border marketplace: state machine, settlement split, idempotent writes, and merchant async notifications.
- Prevented oversell and duplicate orders under flash-sale traffic with MQ decoupling, sharding, distributed locks, and rate limits; stabilized peaks via hot-cache and slow-SQL tuning across multi-region deploy.

Java Microservices MySQL Redis Apache Kafka



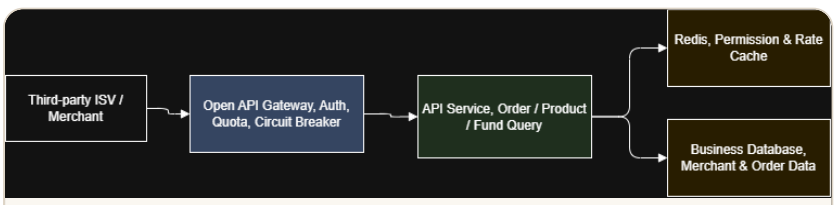
Architecture overview

Merchant Open API Gateway

Backend Software Engineer · AliExpress / Alibaba

- Built merchant open APIs for third-party ISVs to batch-query orders, products, and funds behind auth, permission, quota, and tracing.
- Isolated open-API traffic from core commerce with throttling and graceful degradation so partner integrations stayed reliable without hurting checkout.

Go API gateway Microservices Redis Performance optimization



Gateway architecture

PERSONAL PROJECTS

Personal projects. They do not use or disclose any former employer's confidential business information. Private repositories: access can be granted on request. Public repositories are linked below.

NPC2

Motive-driven generative agents · personal project

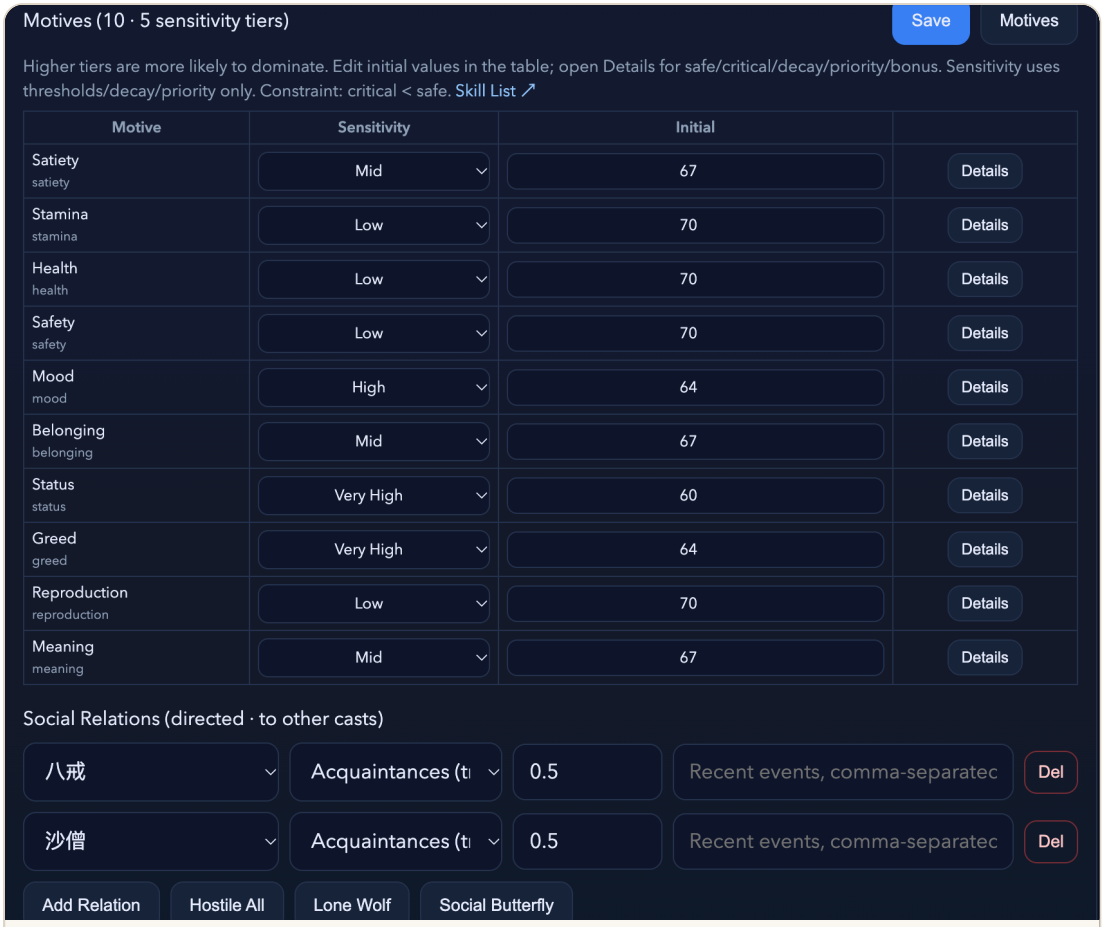
github.com/jeff2025white/npc2 Demo video Private

- Built a sandbox where agents act from competing motives and skills, not fixed scripts; routine pathfinding stays local, while dialogue, conflict, and decisions call an LLM router.
- Shipped a web console to edit motives, relations, and skills and assign different models per character; demo video available, private repo on request.

Python Django Phaser LLM routing



Live sandbox



Motive editor

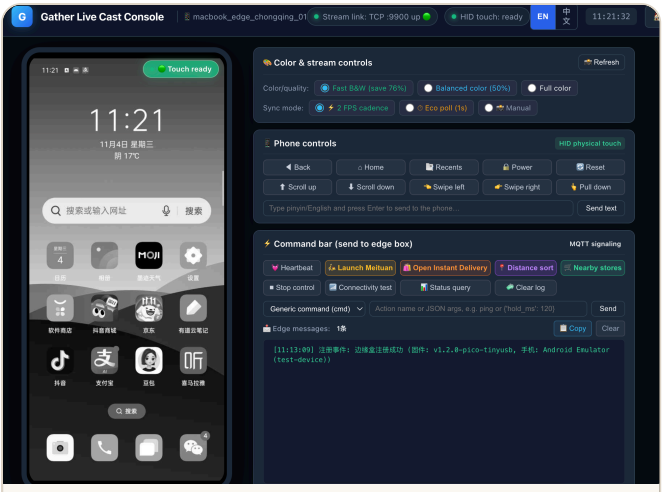
Gather + Flash

In-store capture and quick-commerce retail insight · personal project

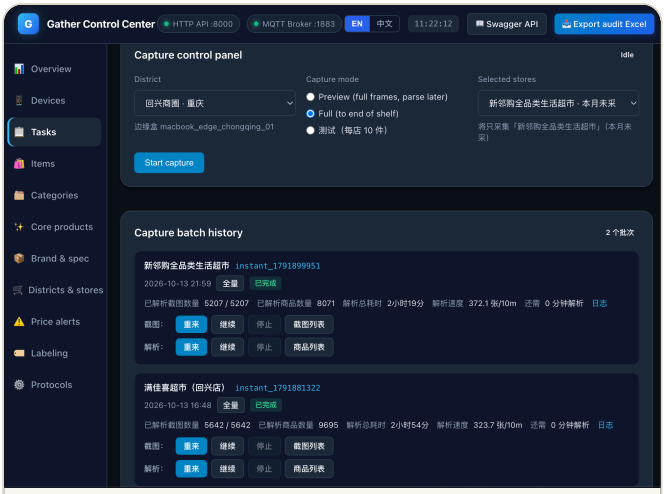
github.com/jeff2025white/gather github.com/jeff2025white/flash Private

- Built an edge capture system that drives a mirrored phone (frames in, HID touch out) to scrape in-store shelf cards for quick-commerce pricing and sales intel.
- Control center batches capture and OCR into price, sales, brand, and spec; Flash turns that catalog into district assortment, price, and sales boards.
- One batch parsed 5,207 screenshots in 2 h 19 min, about 370 frames every 10 minutes.

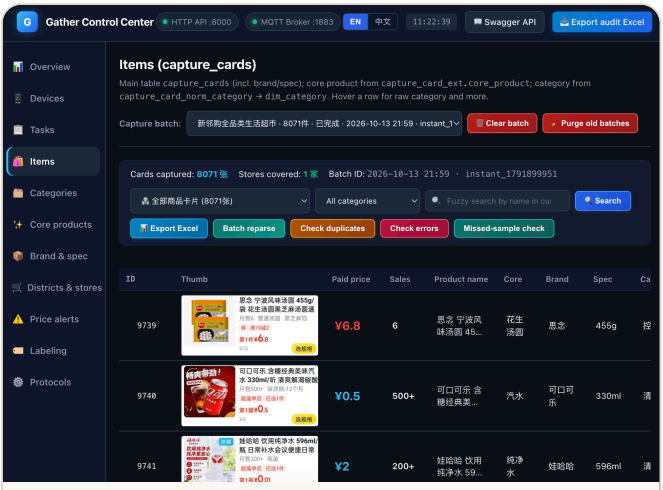
Python FastAPI PostgreSQL SQLite PaddleOCR MQTT RP2040 Kotlin



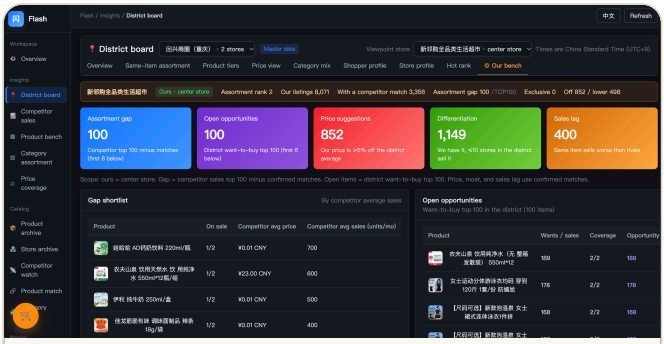
Gather · live cast



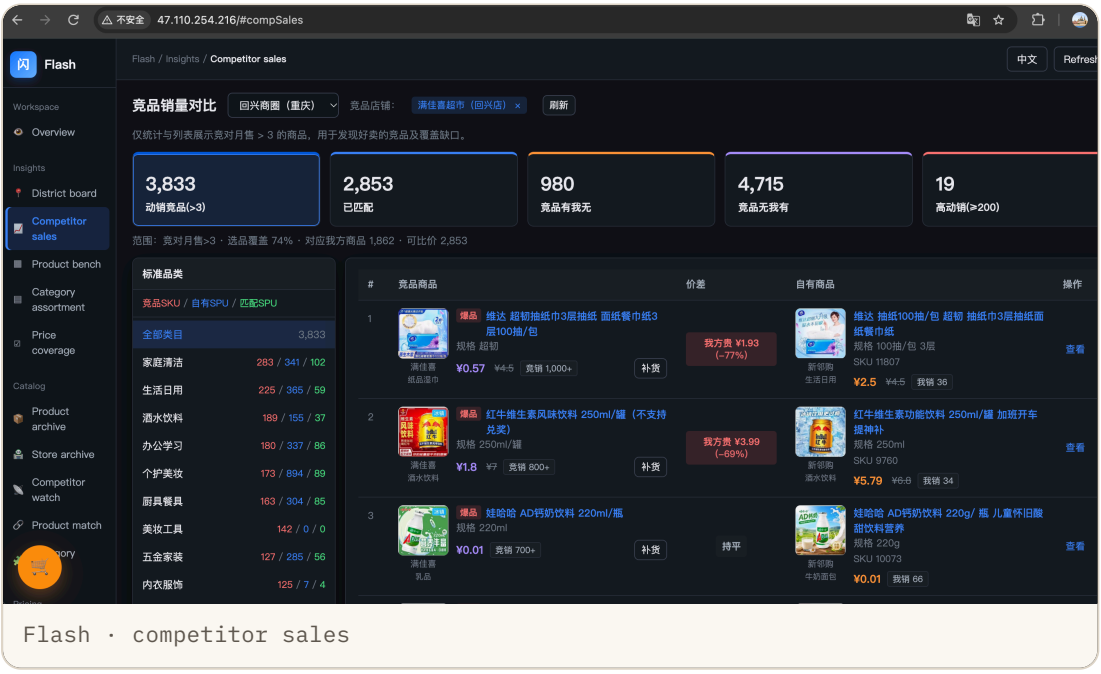
Gather · capture tasks



Gather · parsed cards



Flash · district board



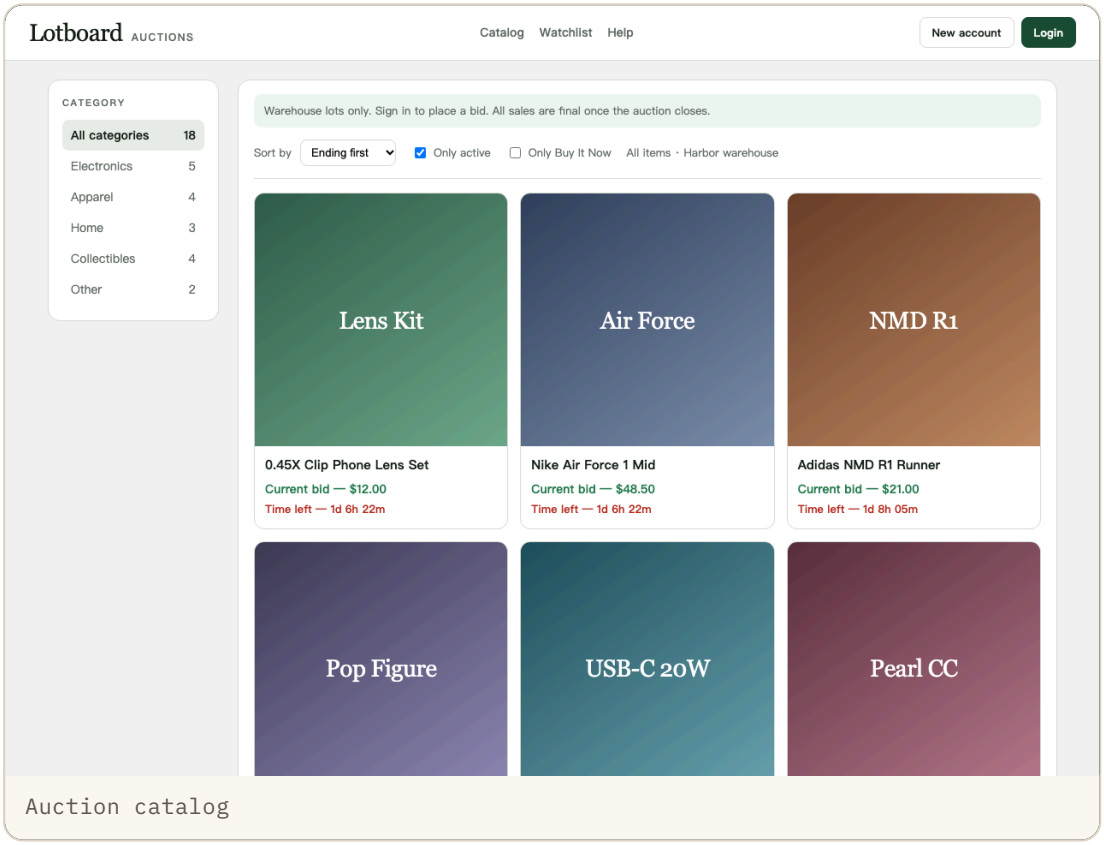
Flash · competitor sales

Online Auction Platform

Production-oriented timed bidding marketplace · personal project

- Built a timed auction marketplace (catalog, bidding, Buy It Now, history) with auth-gated bids.
- Made concurrent bidding safe with transactions and optimistic locking plus bid-increment rules; a scheduler closes lots and picks winners.
- Hardened for production with CSRF/policies, rate-limited bid APIs, Redis cache/queues, and role-based admin.

`Laravel PHP MySQL Redis Queues Scheduler Blade`



Auction catalog

Live Streaming Android Client

Multi-module live / short-video / social app · personal project

github.com/jeff2025white/live-prj-h5-android-p Public

- Shipped a multi-module Android live/social client with login, live rooms, short video, feed, messaging, and wallet against a REST backend.
- Live rooms combine RTC and IM for gifts, danmaku, and DMs; open/close live and room state stay in sync with backend APIs and tokens.
- Modularized with ARouter and product flavors; integrated pay, push, maps, and short-video upload for a releasable APK.

Android Java REST API OkHttp / OkGo ARouter Agora RTC Tencent IM